

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION

Course Code:	CSA16	Course Title:	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Technical Presentation
Weightage:	10% of CIA	Total Marks:	100 (2 Question × 100 Marks)
CO1 Statement:	Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2		
Student Name:	Harish Kumar.M	Reg. No.:	192565053
Section / Batch:	2025	Date:	15.07.2026

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

Question Type	Focus Area	Questions	Marks
Technical Presentation	Decision support system, Operational versus Decision support System, Architecture of data warehouse, ETL, OLAP and data cubes, Dimensional data modelling -star snowflake schemas. Query Processing, Open-source Business Intelligence Tools. Develop an application to perform OLAP operations - Introduction to KNIME tool for reporting.	Q1, Q2	100
GRAND TOTAL:			100 Marks

SECTION A – Technical Presentation

Topic 1: Decision Support Systems (DSS)

☉ Focus Areas:

- Definition and need of DSS
- Components (Data, Model, UI)
- Types of DSS
- Real-world applications (banking, healthcare)
- Advantages & limitations

Topic 2: Operational Systems vs DSS

☉ Focus Areas:

- OLTP vs OLAP concepts
- Key differences (data, users, purpose)
- Performance comparison
- Use-case scenarios
- Integration between systems

Topic 3: Architecture of Data Warehouse

☉ Focus Areas:

- Data warehouse definition
- Three-tier architecture
- ETL process
- Data marts & metadata
- Benefits in decision-making

Topic 4: OLAP and Data Cubes

☉ Focus Areas:

- OLAP concept
- Data cube structure
- OLAP operations (slice, dice, roll-up, drill-down)
- Applications in analytics

Topic 5: Dimensional Data Modeling

☉ Focus Areas:

- Fact and dimension tables
- Star schema vs snowflake schema
- Schema design process
- Real-world example

Topic 6: Query Processing & BI Tools

☉ Focus Areas:

- Query processing stages
- Query optimization
- BI tools overview
- Reporting and visualization
- Examples of open-source BI tools

Topic 7: OLAP Application using KNIME

☉ Focus Areas:

- Introduction to KNIME
- Workflow components (nodes, connections)
- Performing OLAP operations
- Sample application/demo
- Report generation

Code of Conduct

Harish Kumar / 192065052 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Harish Kumar

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Marks Evaluation:

Question No.	Technical Content Accuracy (30%)	Problem Identification (25%)	Use of Tools (20%)	Communication (15%)	Professionalism (10%)	Total Marks (100)
Q1						
Q2						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____



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Ex: 13.025, 13.01

Monday,

Not a Cap

T-03-06

Best suited for	Reporting and BI dashboards	Large systems prioritizing storage efficiency and integrity
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3. Schema Design Process

- Select the business process to model (e.g., order fulfillment)
- Declare the grain — what one fact row represents (e.g., one line item per order)
- Identify dimensions (customer, product, date, store)
- Identify facts / measures (revenue, quantity, discount)
- Build and validate the schema against real query and reporting needs

4. Real-World Example

A retail sales data warehouse illustrates these concepts in practice:

- **Fact_Sales** — grain is one row per product sold per transaction
- Connects to Dim_Product, Dim_Customer, Dim_Store, and Dim_Date

This design lets analysts quickly answer questions like “total revenue by region last quarter” through simple joins, without scanning raw transactional logs.

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